

Question No:6

DS	Session	Tree 1	Question Information	Level 1	Challenge 66
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Question description

A company has n employees, who form a tree hierarchy where each employee has a boss, except for the general director. Your task is to process q queries of the form: who is employee x 's boss k levels higher up in the hierarchy?

Input

The first input line has two integers n and q : the number of employees and queries. The employees are numbered $1, 2, \dots, n$, and employee 1 is the general director. The next line has $n-1$ integers $e_2, e_3, \dots, e_n$: for each employee $2, 3, \dots, n$ their boss. Finally, there are q lines describing the queries. Each line has two integers x and k : who is employee x 's boss k levels higher up?

Output

Print the answer for each query. If such a boss does not exist, print -1 .

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq e_i \leq i-1$
- $1 \leq x \leq n$
- $1 \leq k \leq n$

Logical Test Cases

Test Case 1

INPUT (STDIN)

5 3
1 3 2 2
4 2
1 2

EXPECTED OUTPUT

1
1

Test Case 2

INPUT (STDIN)

5 3
1 1 3 3
4 1
4 2

EXPECTED OUTPUT

3
1

Mandatory Test Cases

Test Case 1

KEYWORD

void link(int i,int j)

Code:

```
1  #include <stdio.h>
2  #define MAXN 200005
3  #define LOG 20
4
5  int up[MAXN][LOG];
6
7  void link(int i, int j) {
8      up[i][0] = j;
9
10     for (int k = 1; k < LOG; k++) {
11         if (up[i][k - 1] != 0)
12             up[i][k] = up[up[i][k - 1]][k - 1];
13         else
14             up[i][k] = 0;
15     }
16 }
17
18 int main() {
19     int n, q;
20
21     scanf("%d %d", &n, &q);
22
23     up[1][0] = 0;
24
25     for (int i = 2; i <= n; i++) {
26         int parent;
27         scanf("%d", &parent);
```

```

29         link(i, parent);
30     }
31
32     while (q--) {
33         int x, k;
34
35         scanf("%d %d", &x, &k);
36
37         for (int j = 0; j < LOG; j++) {
38             if (k & (1 << j)) {
39                 x = up[x][j];
40
41                 if (x == 0)
42                     break;
43             }
44         }
45
46         if (x == 0)
47             printf("-1\n");
48         else
49             printf("%d\n", x);
50     }
51     return 0;
52 }

```

Output:
Test Case:1

```

5 3
1 3 2 2
4 2
1
1 2
-1
1 3
-1

```

Test Case:2

	5 3	
	1 1 3 3	
	4 1	
	3	
	4 2	
	1	
	4 3	
	-1	

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